浙江大學
ZHEJIANG UNIVERSITY

Signal and System
Lab Report

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1 Problem1

1.1 Problem

Problem 1

Consider the following three systems:

System 1: $w[n] = x[n] - x[n-1] - x[n-2]$,
System 2: $y[n] = \cos(x[n])$,
System 3: $z[n] = n\,x[n]$,

where $x[n]$ is the input to each system, and $w[n]$, $y[n]$, and $z[n]$ are the corresponding outputs.

(a). Consider the three inputs signals $x_1[n] = \delta[n]$, $x_2[n] = \delta[n-1]$, and $x_3[n] = (\delta[n] + 2\,\delta[n-1])$. For System 1, store in `w1`, `w2`, and `w3` the responses to the three inputs. The vectors `w1`, `w2`, and `w3` need to contain the values of $w[n]$ only on the interval $0 \le n \le 5$. Use `subplot` and `stem` to plot the four functions represented by `w1`, `w2`, `w3`, and `w1+2*w2` within a single figure. Make analogous plots for Systems 2 and 3.

(b). State whether or not each system is linear. If it is linear, justify your answer. If it is not linear, use the signals plotted in Part (a) to supply a counter-example.

(c). State whether or not each system is time-invariant. If it is time-invariant, justify your answer. If it is not time-invariant, use the signals plotted in Part (a) to supply a counter-example.

Figure 1: Problem1

1.2 Answer

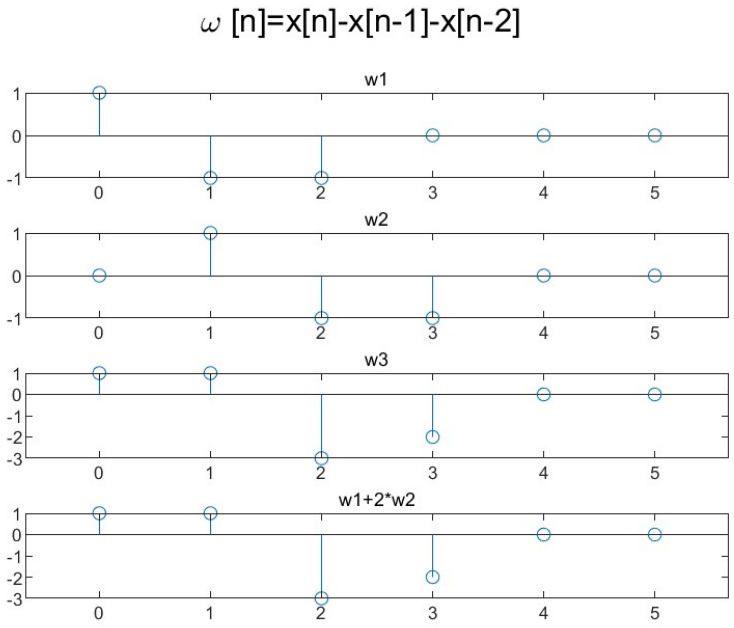


Figure 2: Answer1

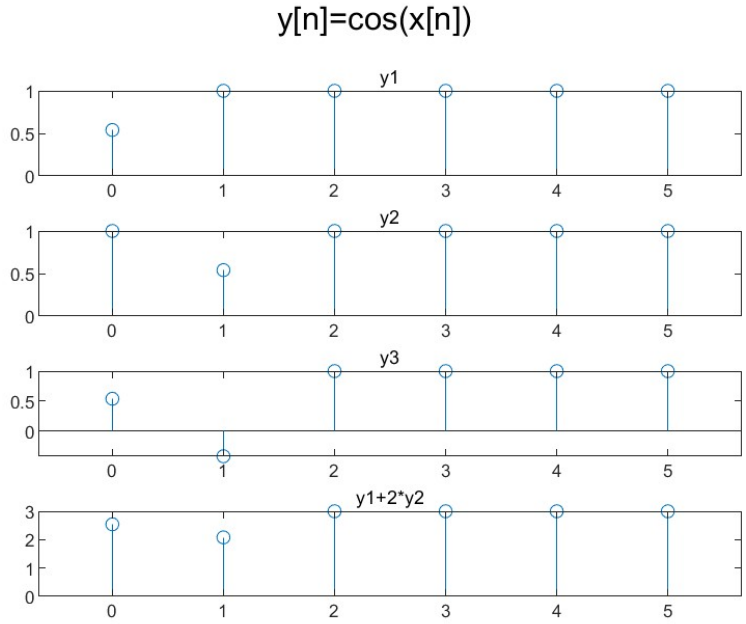


Figure 3: Answer1

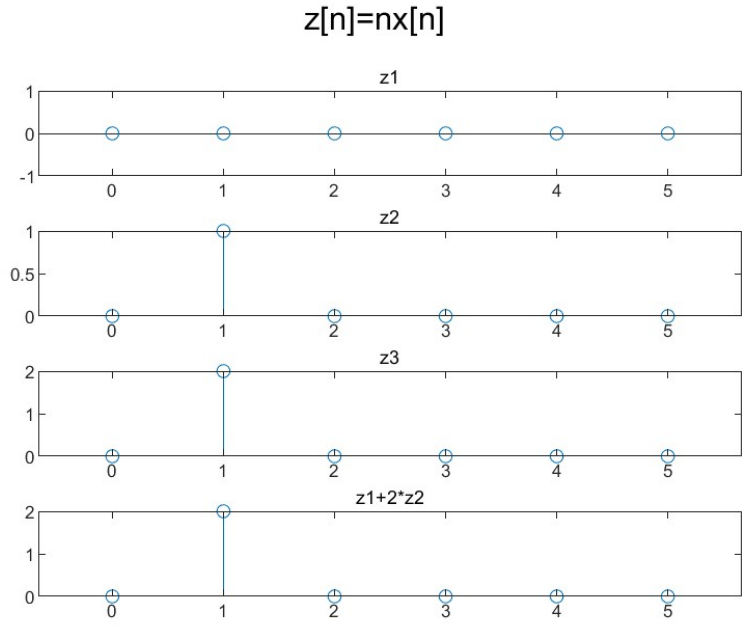


Figure 4: Answer1

(b) Linearity Analysis

A system is **linear** if it satisfies both **additivity** and **homogeneity**. That is, for any inputs $x_1[n]$, $x_2[n]$, and scalars a, b:

$$T\{ax_1[n] + bx_2[n]\} = aT\{x_1[n]\} + bT\{x_2[n]\}$$

System 1: $w[n] = x[n] - x[n-1] - x[n-2]$

- **Homogeneity:** If the input is scaled by a, i.e., $x[n] \to ax[n]$, then

$$w[n] = ax[n] - ax[n-1] - ax[n-2] = a\left(x[n] - x[n-1] - x[n-2]\right)$$

  The output is also scaled by a. Therefore, the system satisfies homogeneity.

- **Additivity:** For two inputs $x_1[n]$ and $x_2[n]$, the outputs are:

$$w_1[n] = x_1[n] - x_1[n-1] - x_1[n-2]$$

$$w_2[n] = x_2[n] - x_2[n-1] - x_2[n-2]$$

  If the input is $x_1[n] + x_2[n]$, then

$$w[n] = (x_1[n] + x_2[n]) - (x_1[n-1] + x_2[n-1]) - (x_1[n-2] + x_2[n-2])$$

$$= w_1[n] + w_2[n]$$

  The system satisfies additivity.

Conclusion: System 1 satisfies both additivity and homogeneity, so it is **linear**.

System 2: $y[n] = \cos(x[n])$

- **Homogeneity:** If the input is scaled by a,

$$y[n] = \cos(ax[n])$$

  In general,

$$\cos(ax[n])ca\cos(x[n])$$

  so the system does not satisfy homogeneity.

- **Additivity:** For two inputs $x_1[n]$ and $x_2[n]$,

$$y_1[n] = \cos(x_1[n]), \quad y_2[n] = \cos(x_2[n])$$

If the input is $x_1[n] + x_2[n]$,

$$y[n] = \cos(x_1[n] + x_2[n]) \neq \cos(x_1[n]) + \cos(x_2[n])$$

so the system does not satisfy additivity.

Also,in Figure3,we can see that $y_3 \neq y_1 + 2 * y_2$.

Conclusion: System 2 fails both additivity and homogeneity, so it is **nonlinear**.

System 3: $z[n] = nx[n]$

- **Homogeneity:** If the input is scaled by a,

$$z[n] = n(ax[n]) = a(nx[n])$$

so the system satisfies homogeneity.

- **Additivity:** For two inputs $x_1[n]$ and $x_2[n]$,

$$z_1[n] = nx_1[n], \quad z_2[n] = nx_2[n]$$

If the input is $x_1[n] + x_2[n]$,

$$z[n] = n(x_1[n] + x_2[n]) = nx_1[n] + nx_2[n] = z_1[n] + z_2[n]$$

so the system satisfies additivity.

Conclusion: System 3 satisfies both additivity and homogeneity, so it is **linear**.

(c) Time-Invariance Analysis

A system is **time-invariant** if shifting the input by k results in an identical shift in the output:

$$T\{x[n-k]\} = y[n-k]$$

System 1: $w[n] = x[n] - x[n-1] - x[n-2]$

- Shift the input by k:

$$x[n] \to x[n-k]$$

The output becomes

$$w[n] = x[n-k] - x[n-k-1] - x[n-k-2]$$

This is equivalent to

$$w[n-k] = x[(n-k)] - x[(n-k)-1] - x[(n-k)-2]$$

so the system is time-invariant.

Conclusion: System 1 is **time-invariant**.

System 2: $y[n] = \cos(x[n])$

- Shift the input by k:

$$x[n] \to x[n-k]$$

The output becomes

$$y[n] = \cos(x[n-k])$$

This is equivalent to

$$y[n-k] = \cos(x[(n-k)])$$

so the system is time-invariant.

Conclusion: System 2 is **time-invariant**.

System 3: $z[n] = nx[n]$

- Shift the input by k:

$$x[n] \to x[n-k]$$

The output becomes

$$z[n] = n \cdot x[n-k]$$

The expected time-invariant output should be

$$z[n-k] = (n-k) \cdot x[n-k]$$

but

$$nx[n-k] \neq (n-k)x[n-k]$$

so the system is time-variant.

Additionally, in Figure 4, we can see that z_1 shifted 1 unit to the right is not equal to z_2.

Conclusion: System 3 is **time-variant**.

Final Summary

System	Linearity	Time-Invariance
System 1	Linear	Time-Invariant
System 2	Nonlinear	Time-Invariant
System 3	Linear	Time-Variant

2 Problem2a,2b

2.1 Problem

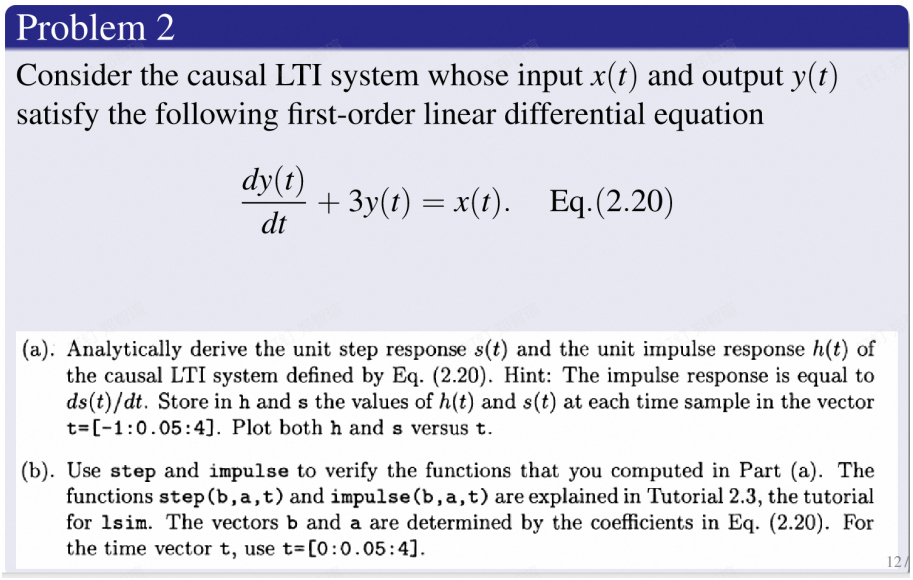


Figure 5: Problem2a,2b

2.2 Answer

(a) Analytical Derivation of the Step Response $s(t)$ and Impulse Response $h(t)$

Consider the causal LTI system defined by the differential equation:

$$\frac{dy(t)}{dt} + 3y(t) = x(t) \qquad \text{Eq. (2.20)}$$

Step 1: Laplace Transform of the Equation

Taking the Laplace Transform of both sides of the differential equation (assuming zero initial conditions, i.e., $y(0^-) = 0$):

$$s \cdot Y(s) + 3Y(s) = X(s)$$

Factor out $Y(s)$:

$$Y(s)\,(s+3) = X(s)$$

Thus, the transfer function $H(s)$ is:

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s+3}$$

Step 2: Impulse Response $h(t)$

The impulse response $h(t)$ is the inverse Laplace Transform of $H(s)$:

$$h(t) = \mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\}$$

We have:

$$h(t) = e^{-3t}u(t)$$

where $u(t)$ is the unit step function.

Step 3: Step Response $s(t)$

The step response $s(t)$ is the system's response to a unit step input $x(t) = u(t)$.

In the Laplace domain, the input is:

$$X(s) = \frac{1}{s}$$

Then the output is:

$$S(s) = H(s) \cdot X(s) = \frac{1}{s+3} \cdot \frac{1}{s} = \frac{1}{s(s+3)}$$

Perform partial fraction decomposition:

$$\frac{1}{s(s+3)} = \frac{A}{s} + \frac{B}{s+3}$$

Multiply both sides by $s(s+3)$:

$$1 = A(s+3) + B(s)$$

Solving for A and B:

- Let $s = 0$: $1 = A(0+3) \Rightarrow A = \frac{1}{3}$
- Let $s = -3$: $1 = B(-3) \Rightarrow B = -\frac{1}{3}$

Thus:

$$S(s) = \frac{1}{3} \cdot \frac{1}{s} - \frac{1}{3} \cdot \frac{1}{s+3}$$

Taking the inverse Laplace Transform:

$$s(t) = \frac{1}{3}\left[1 - e^{-3t}\right] u(t)$$

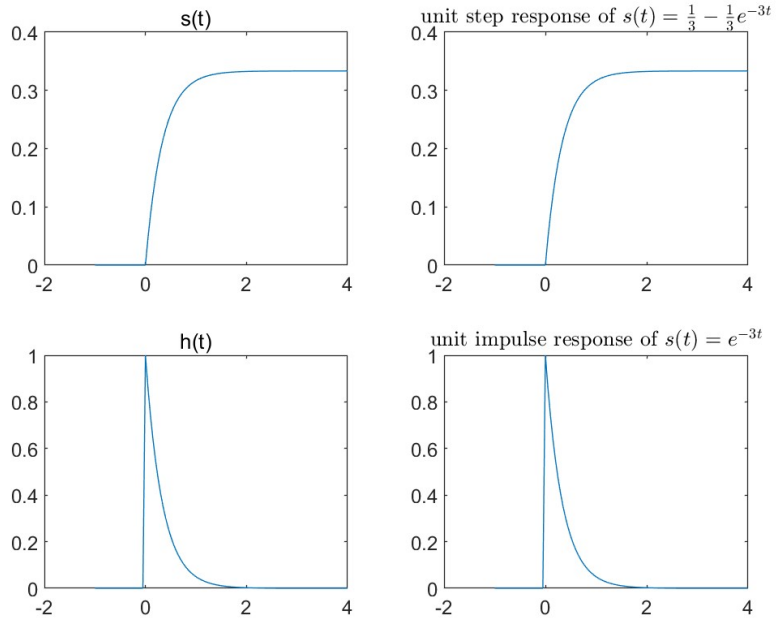


Figure 6: Answer2b

3 Problem2c,2d

3.1 Problem

Problem 2

(c). Use `lsim` to simulate the response to $\delta^{\Delta}(t)$ for $\Delta = 0.1$, 0.2, and 0.4. For the time vector, use `t=[-1:0.05:4]`, and store in `d1`, `d2`, and `d3` the corresponding values of $\delta^{\Delta}(t)$ for $\Delta = 0.1$, 0.2, and 0.4, respectively. Remember that $\delta^{\Delta}(\Delta) = 0$, not $1/\Delta$. Store in `h1`, `h2`, and `h3` the simulated response for $\Delta = 0.1$, 0.2, and 0.4, respectively. These vectors contain the simulated values of $h^{\Delta}(t)$ at the time samples given in `t`.

On three separate figures, plot the simulated values of $h^{\Delta}(t)$. On each figure, also include a plot of $h(t)$, which was stored in `h` in Part (a). How does $h^{\Delta}(t)$ compare to $h(t)$ as Δ decreases?

(d). Why is it necessary to ensure that the pulse $\delta^{\Delta}(t)$ has unit area? Namely, what if the response to

$$D^{\Delta}(t) = \begin{cases} 1, & 0 \le t < \Delta, \\ 0, & \text{otherwise}, \end{cases}$$

13/

Figure 7: Problem2c,2d

3.2 Answer

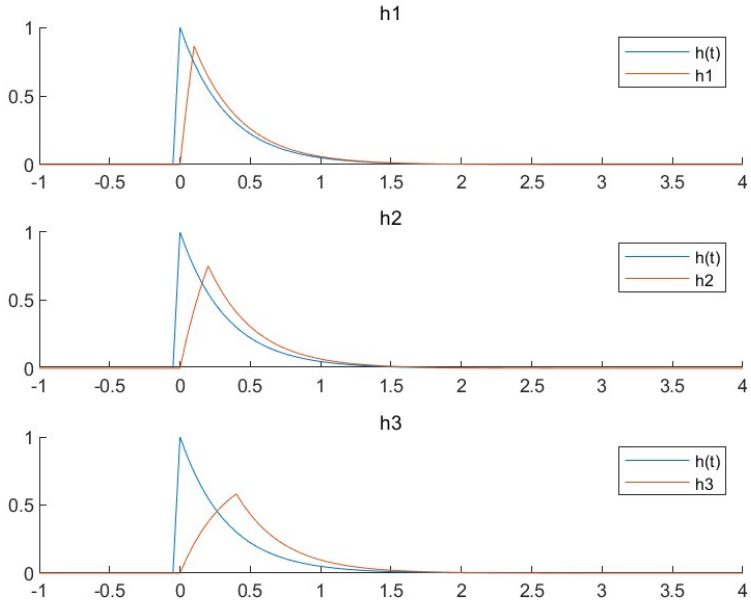


Figure 8: Answer2c

Answer2d : If we directly use $D^{\Delta}(t)$ in place of $\delta^{\Delta}(t)$, the system's response will be scaled by Δ. This means that if Δ is small, the response will be underestimated, and if Δ is large, it will be overestimated.

4 Problem2e

4.1 Problem

Problem 2

(e). Now consider the following function

$$d_a(t) = a\, e^{-at}\, u(t)\,.$$

for $a > 0$. While $d_a(t)$ has infinite duration, most of the signal energy is contained in the interval $0 \le t \le 4/a$. For sufficiently large values of a, the response of the LTI system to $d_a(t)$ will for all practical purposes be identical to $h(t)$. What is the area of $d_a(t)$ as a function of $a > 0$?

Use `t=[-1:0.05:4]`, and store in `da1`, `da2`, and `da3` the corresponding values of $d_a(t)$ for $a = 4$, 8, and 16, respectively. For each of the three values of a, use `lsim` to simulate the response of the LTI system to $d_a(t)$. Plot each function on a separate figure using `plot`. On each figure, also include a plot of the impulse response computed in Part (a). How does the response to $d_a(t)$ compare to $h(t)$ for large values of a?

Figure 9: Problem2e

4.2 Answer

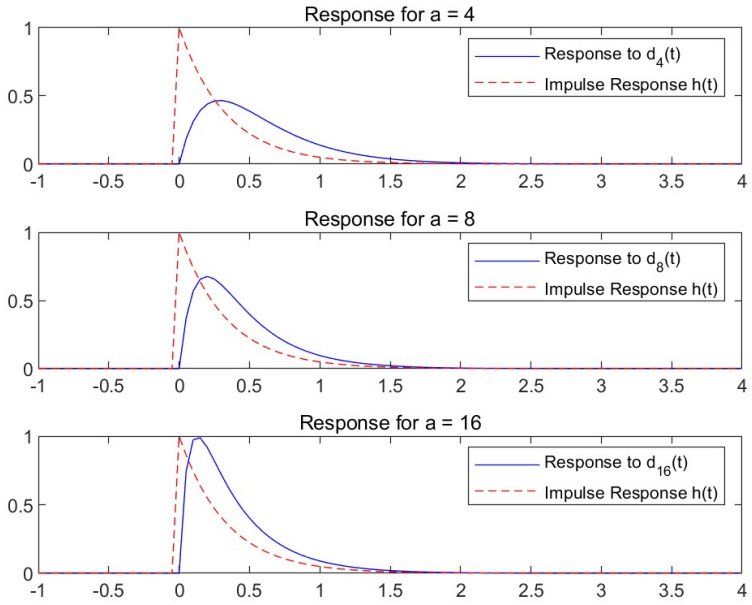


Figure 10: Answer2e

For large a, the response to $d_a(t)$ approximates the impulse response $h(t)$.

5 Problem3

5.1 Problem

Problem 3

In this exercise, you will consider the problem of removing an echo from a recording of a speech signal. This project will use the audio capabilities of MATLAB to play recordings of both the original speech and the result of your processing. To begin this exercise you will need to load the speech file `lineup.mat`, which is contained in the Computer Explorations Toolbox. The Computer Explorations Toolbox can be obtained from The MathWorks at the address provided in the Preface. If this speech file is already somewhere in your MATLABPATH, then you can load the data into MATLAB by typing

```
>> load lineup.mat
```

You can check your MATLABPATH, which is a list of all the directories which are currently accessible by MATLAB, by typing `path`. The file `lineup.mat` must be in one of these directories.

Once you have loaded the data into MATLAB, the speech waveform will be stored in the variable `y`. Since the speech was recorded with a sampling rate of 8192 Hz, you can hear the speech by typing

```
>> sound(y,8192)
```

Figure 11: Problem3

Problem 3

You should hear the phrase "line up" with an echo. The signal $y[n]$, represented by the vector `y`, is of the form

$$y[n] = x[n] + \alpha x[n - N], \tag{2.21}$$

where $x[n]$ is the uncorrupted speech signal, which has been delayed by N samples and added back in with its amplitude decreased by $\alpha < 1$. This is a reasonable model for an echo resulting from the signal reflecting off of an absorbing surface like a wall. If a microphone is placed in the center of a room, and a person is speaking at one end of the room, the recording will contain the speech which travels directly to the microphone, as well as an echo which traveled across the room, reflected off of the far wall, and then into the microphone. Since the echo must travel further, it will be delayed in time. Also, since the speech is partially absorbed by the wall, it will be decreased in amplitude. For simplicity ignore any further reflections or other sources of echo.

For the problems in this exercise, you will use the value of the echo time, $N = 1000$, and the echo amplitude, $\alpha = 0.5$.

Basic Problems

(a). In this exercise you will remove the echo by linear filtering. Since the echo can be represented by a linear system of the form Eq. (2.21), determine and plot the impulse

Figure 12: Problem3

Problem 3

response of the echo system Eq. (2.21). Store this impulse response in the vector **he** for $0 \leq n \leq 1000$.

(b). Consider an echo removal system described by the difference equation

$$z[n] + \alpha z[n - N] = y[n], \tag{2.22}$$

where $y[n]$ is the input and $z[n]$ is the output which has the echo removed. Show that Eq. (2.22) is indeed an inverse of Eq. (2.21) by deriving the overall difference equation relating $z[n]$ to $x[n]$. Is $z[n] = x[n]$ a valid solution to the overall difference equation?

Intermediate Problems

(c). The echo removal system Eq. (2.22) will have an infinite-length impulse response. Assuming that $N = 1000$, and $\alpha = 0.5$, compute the impulse response using `filter` with an input that is an impulse given by `d=[1 zeros(1,4000)]`. Store this 4001 sample approximation to the impulse response in the vector **her**.

(d). Implement the echo removal system using `z=filter(1,a,y)`, where **a** is the appropriate coefficient vector derived from Eq. (2.22). Plot the output using `plot`. Also, listen to the output using `sound`. You should no longer hear the echo.

Figure 13: Problem3

5.2 Answer

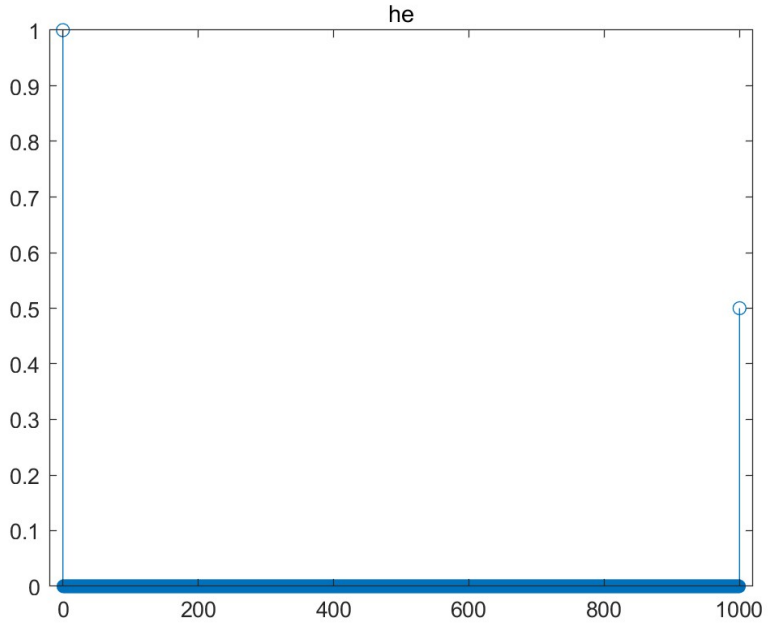


Figure 14: Answer3a

Answer3b : To prove the echo removal system is the inverse, substitute $y[n] = x[n] + \alpha x[n - N]$ into $z[n] + \alpha z[n - N] = y[n]$. Assuming $z[n] = x[n]$, both sides match, confirming $z[n] = x[n]$ is valid. Thus, the system is indeed the inverse. so $z[n] = x[n]$ is the valid solution.

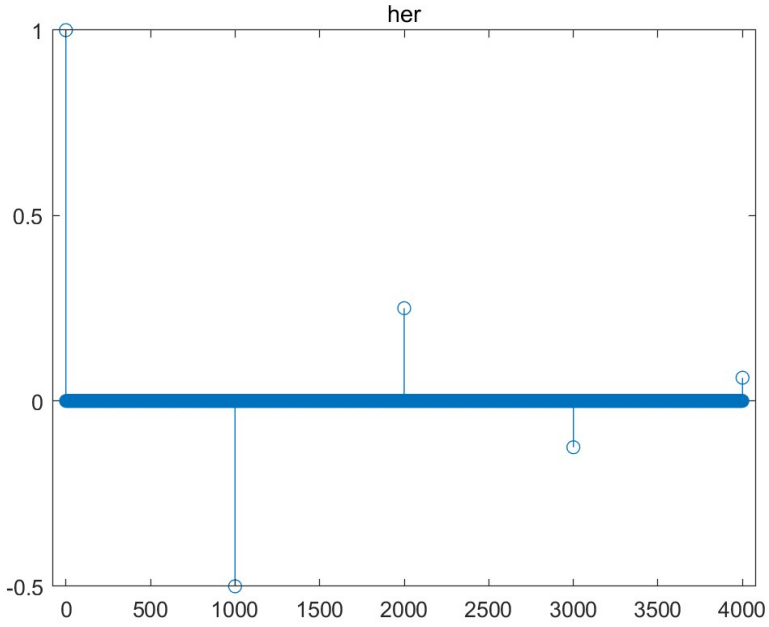


Figure 15: Answer3c

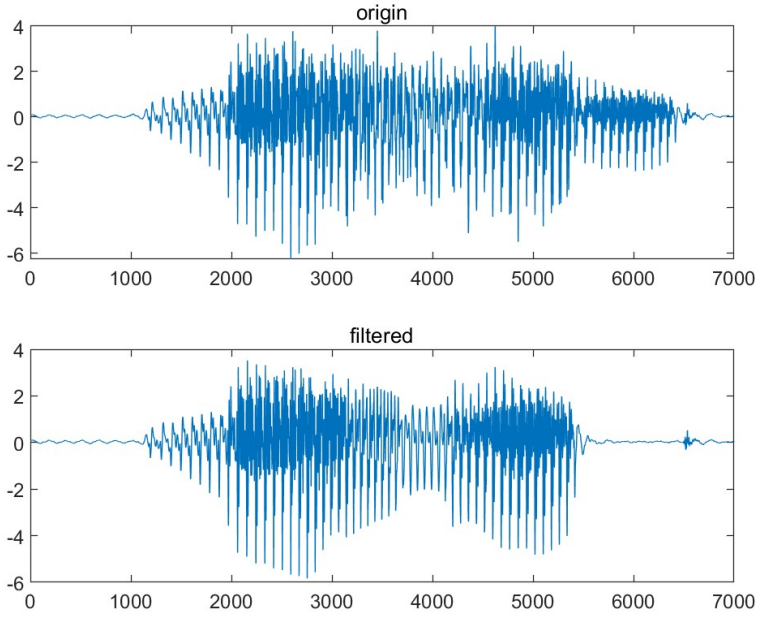


Figure 16: Answer3d

6 Code

6.1 Code1

Listing 1: Problem1a Code1

```
clear;
clc;

n = 0:5;
x1 = (n == 0);
x2 = (n == 1);
x3 = (n == 0) + 2*(n == 1);

a=1;
b=[1 -1 -1];

w1=filter(b,a,x1);
w2=filter(b,a,x2);
w3=filter(b,a,x3);
w4=w1+2*w2;

subplot(4,1,1);
stem(n,w1);
title("w1");

subplot(4,1,2);
stem(n,w2);
title("w2");

subplot(4,1,3);
stem(n,w3);
title("w3");

subplot(4,1,4);
stem(n,w4);
title("w1+2*w2");

sgtitle("\omega [n]=x[n]-x[n-1]-x[n-2]");
```

Listing 2: Problem1a Code2

```
clear;
clc;

n=0:5;
x1=[1 0 0 0 0 0];
x2=[0 1 0 0 0 0];
x3=x1+2*x2;

w1=cos(x1);
w2=cos(x2);
w3=cos(x3);
w4=w1+2*w2;


subplot(4,1,1);
stem(n,w1);
title('y1');

subplot(4,1,2);
stem(n,w2);
title('y2');

subplot(4,1,3);
stem(n,w3);
title('y3');

subplot(4,1,4);
stem(n,w4);
title('y1+2*y2');

sgtitle("y[n]=cos(x[n])");
```

Listing 3: Problem1a Code3

```
clear;
clc;

n=0:5;
x1=[1 0 0 0 0 0];
x2=[0 1 0 0 0 0];
x3=x1+2*x2;

w1=n.*x1;
w2=n.*x2;
w3=n.*x3;
w4=w1+2*w2;


subplot(4,1,1);
stem(n,w1);
title('z1');

subplot(4,1,2);
stem(n,w2);
title('z2');

subplot(4,1,3);
stem(n,w3);
title('z3');

subplot(4,1,4);
stem(n,w4);
title('z1+2*z2');

sgtitle("z[n]=nx[n]");
```

6.2 Code2

Listing 4: Problem2b Code

```
clear;
clc;

t=-1:0.05:4;
t_=0:0.05:4;
u=[zeros(1,length(-1:0.05:-0.05)) ones(1,length(0:0.05:4))];

s=(1/3-1/3*exp(-3*t)).*u;
h=exp(-3*t).*u;

a=[1 3];
b=1;

s1=step(b,a,t_);
s1=[zeros(1,length(-1:0.05:-0.05)) s1.'];

h1=impulse(b,a,t_);
h1=[zeros(1,length(-1:0.05:-0.05)) h1.'];

subplot(2,2,2);
plot(t,s);
title('$\frac{1}{3} - \frac{1}{3} e^{-3t}$', 'Interpreter', 'latex')

subplot(2,2,1);
plot(t,s1);
title('s(t)');

subplot(2,2,4);
plot(t,h);
title('$e^{-3t}$', 'Interpreter', 'latex')

subplot(2,2,3);
plot(t,h1);
title('h(t)');
```

Listing 5: Problem2c Code

```
clear;
clc;

t=-1:0.05:4;
t_=0:0.05:4;

delta1=0.1;
delta2=0.2;
delta3=0.4;

d1=[zeros(1,length(-1:0.05:-0.05)) ones(1,length(0:0.05:delta1-0.05))/
   delta1 zeros(1,length(delta1:0.05:4))];
d2=[zeros(1,length(-1:0.05:-0.05)) ones(1,length(0:0.05:delta2-0.05))/
   delta2 zeros(1,length(delta2:0.05:4))];
d3=[zeros(1,length(-1:0.05:-0.05)) ones(1,length(0:0.05:delta3-0.05))/
   delta3 zeros(1,length(delta3:0.05:4))];

a=[1 3];
b=1;
h1=lsim(b,a,d1,t);
h2=lsim(b,a,d2,t);
h3=lsim(b,a,d3,t);
h=impulse(b,a,t_);
h=[zeros(1,length(-1:0.05:-0.05)) h.'];

subplot(3,1,1);
hold on;
p1 = plot(t,h, 'DisplayName', 'h(t)');
p2 = plot(t,h1, 'DisplayName', 'h1');
title('h1');
legend([p1, p2]);

subplot(3,1,2);
hold on;
p1 = plot(t,h, 'DisplayName', 'h(t)');
p2 = plot(t,h2, 'DisplayName', 'h2');
title('h2');
```

```
legend([p1, p2]);

subplot(3,1,3);
hold on;
p1 = plot(t,h, 'DisplayName', 'h(t)');
p2 = plot(t,h3, 'DisplayName', 'h3');
title('h3');
legend([p1, p2]);
```

Listing 6: Problem2e Code

```
clear;
clc;

t=-1:0.05:4;

a1 = 4; a2 = 8; a3 = 16;
da1 = a1 * exp(-a1 * t) .* (t >= 0);
da2 = a2 * exp(-a2 * t) .* (t >= 0);
da3 = a3 * exp(-a3 * t) .* (t >= 0);

h = exp(-3 * t) .* (t >= 0);

a=[1 3];
b=1;
y1 = lsim(b, a, da1, t);
y2 = lsim(b, a, da2, t);
y3 = lsim(b, a, da3, t);

subplot(3,1,1);
plot(t, y1, 'b', t, h, 'r--');
legend('Response to d_4(t)', 'Impulse Response h(t)');
title('Response for a = 4');

subplot(3,1,2);
plot(t, y2, 'b', t, h, 'r--');
legend('Response to d_8(t)', 'Impulse Response h(t)');
title('Response for a = 8');

```

```
29  subplot(3,1,3);
30  plot(t, y3, 'b', t, h, 'r--');
31  legend('Response to d_{16}(t)', 'Impulse Response h(t)');
32  title('Response for a = 16');
```

6.3 Code3

Listing 7: Problem3a Code

```
clear;
clc;

alpha=0.5;
N=1000;
n=0:1000;

he = [1 zeros(1,N-1) alpha];

stem(n,he);
title("he");
```

Listing 8: Problem3c Code

```
clear;
clc;

n = 0:4000;
N = 1000;
alpha = 0.5;
a = [1, zeros(1, N-1), alpha];
d = [1, zeros(1, 4000)];
her = filter(1, a, d);
stem(n,her);
title('her');
```

Listing 9: Problem3d Code

```
clear;
clc;

load lineup.mat

alpha=0.5;
n=0:1:7000;
N=1000;

a=1;
b=[1 zeros(1,N-1) alpha];

y_ = filter(a,b,y);
sound(y ,8192);
pause(2);
sound(y_ ,8192);

subplot(2,1,1)
plot(y);
title('origin');

subplot(2,1,2)
plot(y_);
title('filtered')
```